

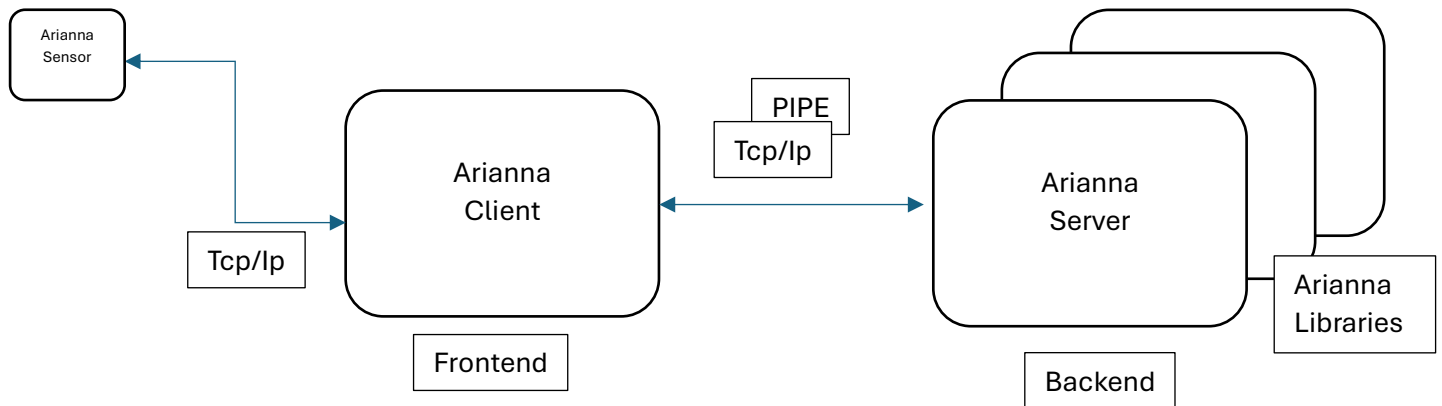
AriannaServer interface

Server interface description

Sommario

1. Introduction.....	3
2. Command description	5
2.1. Init #.....	5
2.2. SetParameters #	5
2.3. Clear #	5
2.4. ClearAll.....	5
2.5. Set #...	5
2.6. Get #	6
2.7. Last #.....	6
3. Examples	7
3.1. Init #.....	7
3.2. SetParameters #	7
3.3. Clear #	7
3.4. ClearAll.....	8
3.5. Set #...	8
3.6. Get #	8
3.7. Last #.....	10
4. Other functionalities	12
4.1. Wayback	12
4.2. Inertial - GPS fusion	12

1. Introduction



The AriannaSensor is an autonomous system that implements internally all the data collection and the algorithms to obtain the relative user track. By using an external communication device, it can transmit the user position to a **C**ommand and **C**ontrol **C**onsole.

There are two options to visualize the user position on the map on the CCC, the AriannaMap software and a custom-developed application. This document gives a preliminary description of the strategy to interact with the Arianna sensor & Software using a custom-developed user interface. Basically, the user interface is the frontend side and the AriannaServer is the backend side of this integration.

While CCC is implemented as a graphical frontend to visualize the user position with or without the full track on a map, the backend performs the relative to absolute transformation and an auxiliary track refinement. The backend is implemented into a stand-alone application (AriannaServer) that can collect the received data and apply the final track refinement.

The AriannaServer is an independent application written in C# and it can be cross compiled for many different operating systems like Windows, Linux, MacOS, IOS, Android.

The backend layer can interact with the graphical frontend on an IPC (**I**nter **P**rocess **C**ommunication) line like a PIPE or by using a TCP/IP connection.

The sequence of operations is:

- The remote Sensor produces the data stream (#...) in the 74h format.
- It sends these data on the BLE connection (PAN) to a smartphone (or directly on the WAN to the CCC by using the Wi-Fi connection).

- If the smartphone App is used, it sends the received data stream to the CCC (or directly to the remote server) on the WAN (Point to Point connection)
- The CCC gets from the AriannaServer the corrected track on request.

The AriannaServer manages the following commands to interact with the CCC:

- Init #
- SetParameter #
- Clear #
- ClearAll
- Set #....
- Get #
- Last #
- Wayback #

2. Command description

2.1. Init

The CCC can use this command to initialize the user (#) information.

2.2. SetParameters

The CCC uses this command to set or update the following parameters of the selected (#) user:

- User ID.
- Offset from the starting point in meters (X).
- Offset from the starting point in meters (Y).
- North orientation correction in radians.
- Track compensation value.
- 0.0 (for internal use)
- Track type selection:
 - 1 => Backward corrected track.
 - 4 => User corrected track.
- Starting point Latitude.
- Starting point Longitude.

2.3. Clear

Using this command the CCC can clear all the collected information for the selected (#) user.

2.4. ClearAll

Using this command the CCC can clear all the collected information for all the users.

2.5. Set #...

With this command the CCC adds a single message to the selected (#) user memory.

2.6. Get

With this command the CCC retrieves the full track of the selected (#) user.

2.7. Last

This command is used to retrieve the last position for the selected (#) user only.

3. Examples

3.1. Init

Init 104

This command Initialize the internal memory for the user 104.

The user Id (#) must be in the range 0...255.

3.2. SetParameters

SetParameters 104,2.5,-3.7,0.034906585039886591,0.1,0,4,41.87892716196967,12.508081927663124

This command sets the internal parameters of the user 104.

The user Id must be in the range 0...255.

- 104 => User ID
- 2.5 => Offset from the starting point in meters (X)
- -3.7 => Offset from the starting point in meters (Y)
- 0.034906585039886591 => North orientation correction in radians (==2°)
- 0.1 => Track compensation value
- 0 => (for internal use, put 0.0)
- 4 => Track type selection (User corrected track)
- 41.87892716196967 => Starting point Latitude
- 12.508081927663124 => Starting point Longitude

3.3. Clear

Clear 104

It clears the internal memory for the user 104.

The user Id (#) must be in the range 0...255.

Example

```
104:0,41.87892716196967,12.508081927663124,0,0,0
104:1,41.87892716196967,12.508081927663124,0,-0.036,0
104:2,41.87892716196967,12.508081927663124,0,-0.105,0
104:3,41.87892716196967,12.508081927663124,0,-0.051999997,0
104:4,41.87892716196967,12.508081927663124,0,-0.152,0
104:5,41.87892716196967,12.508081927663124,0,-0.058999997,0
104:6,41.87892716196967,12.508081927663124,0,-0.047000002,0
104:7,41.87892716196967,12.508081927663124,0,0.008,0
104:8,41.87892716196967,12.508081927663124,0,0.074,0
104:9,41.87892716196967,12.508081927663124,0,0.05,0
104:10,41.87892716196967,12.508081927663124,0,-0.072,0
104:11,41.87892716196967,12.508081927663124,0,-0.012999999,0
104:12,41.87892716196967,12.508081927663124,0,0.125,0
104:13,41.87892716196967,12.508081927663124,0,0.168,0
104:14,41.87892716196967,12.508081927663124,0,0.122,0
104:15,41.87892716196967,12.508081927663124,0,0.044999998,0
104:16,41.87892716196967,12.508081927663124,0,0.001,0
104:17,41.87892716196967,12.508081927663124,0,0.03,0
104:18,41.87892716196967,12.508081927663124,0,-0.003,0
104:19,41.87892716196967,12.508081927663124,0,-0.128,0
104:20,41.87892716196967,12.508081927663124,0,-0.125,0
104:21,41.87892716196967,12.508081927663124,0,-0.096,0
104:22,41.87892716196967,12.508081927663124,0,-0.129,0
104:23,41.87892716196967,12.508081927663124,0,0.006,0
104:24,41.87892716196967,12.508081927663124,0,0.055000003,0
104:25,41.87892716196967,12.508081927663124,0,-0.021,0
104:26,41.87892716196967,12.508081927663124,0,0.001,0
104:27,41.87892716196967,12.508081927663124,0,0.073,0
104:28,41.87892716196967,12.508081927663124,0,-0.042,0
104:29,41.87892716196967,12.508081927663124,0,-0.042999998,0
104:30,41.87892716196967,12.508081927663124,0,-0.072,0
104:31,41.87892716196967,12.508081927663124,0,-0.051999997,0
104:32,41.87892716196967,12.508081927663124,0,0.018,0
104:33,41.87892716196967,12.508081927663124,0,0.004,0
104:34,41.87892716196967,12.508081927663124,0,0.13399999,0
104:35,41.87892716196967,12.508081927663124,0,0.189,0
104:36,41.87892716196967,12.508081927663124,0,0.203,0
104:37,41.87892716196967,12.508081927663124,0,0.074,0
104:38,41.87892716196967,12.508081927663124,0,0.101,0
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104:42,41.87892716196967,12.508081927663124,0,0.224,0
104:43,41.87892716196967,12.508081927663124,0,0.182,0
104:44,41.87892716196967,12.508081927663124,0,0.114,0
104:45,41.87892716196967,12.508081927663124,0,0.15799999,0
104:46,41.87892716196967,12.508081927663124,0,0.048,0
104:47,41.87892716196967,12.508081927663124,0,-0.036,0
104:48,41.87892716196967,12.508081927663124,0,-0.003,0
104:49,41.87892716196967,12.508081927663124,0,-0.05,0
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104:51,41.87892716196967,12.508081927663124,0,0.019,0
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104:56,41.87892716196967,12.508081927663124,0,0.539,0
104:57,41.87892716196967,12.508081927663124,0,1.311,0
104:58,41.87892356468602,12.508077096222308,-0.001,-1.443,-0.001379310390269705
104:59,41.87892356468602,12.508077096222308,-0.001,-2.3500001,-0.001379310390269705
104:60,41.87891367215618,12.50808071980301,0,0.791,-0.0007586206855445072
104:61,41.87891367215618,12.50808071980301,0,1.489,-0.0007586206855445072
104:62,41.878916370118446,12.508096421985933,0.001,0.703,-0.00013793098081930954
104:63,41.878928061290786,12.508096421985933,0.002,-2.232,0.00048275872390588816
104:64,41.878928061290786,12.508096421985933,0.002,-0.96799994,0.00048275872390588816
104:65,41.878937953820895,12.508090382684282,0.004,-5.4199996,0.002103448476128537
104:66,41.87893705449978,12.508081927663124,0.001,-6.558,-0.0012758620091360704
104:67,41.87893705449978,12.508081927663124,0.001,-5.365,-0.0012758620091360704
```

104:68,41.87893705449978,12.508081927663124,0.001,-1.9109999,-0.0012758620091360704
104:69,41.87893345721586,12.50809279840523,0.002,-0.82,-0.0006551723044108727
104:70,41.87892086672348,12.508095214125458,0.001,1.717,-0.0020344826946805776
104:71,41.87891367215618,12.50808675910358,0.001,0.45499998,-0.0020344826946805776
104:72,41.87891367215618,12.50808675910358,0.001,0.368,-0.0020344826946805776
104:73,41.87891816876068,12.50808071980265,0.002,0.12099999,-0.00141379298995538
104:74,41.87891816876068,12.50808071980265,0.002,1.742,-0.00141379298995538
104:75,41.87892356468629,12.508079511942535,0.001,-1.648,-0.002793103380225085
104:76,41.87892356468629,12.508079511942535,0.001,2.4359999,-0.002793103380225085
104:77,41.878922665365444,12.508077096221948,0.001,4.019,-0.002793103380225085
104:78,41.878922665365444,12.508077096221948,0.001,3.197,-0.002793103380225085
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104:81,41.87890737690945,12.508084343383173,0.003,-0.14299999,-0.0011724137444177577
104:82,41.87890108166273,12.508100045564474,0.002,-1.61,-0.0025517240182721408
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104:88,41.87893975246312,12.508139904954357,0.004,0.197,-0.0016896549515939994
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104:90,41.878938853143076,12.50815560713332,0.004,-3.6520002,-0.0016896549515939994
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104:95,41.87892626264963,12.50816527001711,0.004,4.3450003,-0.0024482756371385066
104:96,41.87891277283614,12.508160438575215,0.005,0.71,-0.0018275862816592744
104:97,41.87891277283614,12.508160438575215,0.005,6.014,-0.0018275862816592744
104:98,41.87890557826723,12.508151983556218,0.005,1.336,-0.0018275862816592744
104:99,41.878894786416005,12.508148359973355,0.004,4.121,-0.003206896322683014
104:100,41.87889568573819,12.508166477876145,0.005,-1.3939999,-0.0025862069672037817
104:101,41.87889568573819,12.508166477876145,0.005,-0.58599997,-0.0025862069672037817
104:102,41.87889748438042,12.508184595778935,0.006,-0.042999998,-0.001965517146063262
104:103,41.878901980984914,12.508199090098865,0.005,6.153,-0.003344827652748289
104:104,41.878914571477296,12.508199090098865,0.006,4.6879997,-0.0027241378316077694
104:105,41.878914571477296,12.508199090098865,0.006,1.8699999,-0.0027241378316077694
104:106,41.87892716196967,12.508195466527521,0.006,1.574,-0.0027241378316077694
104:107,41.878938853142,12.508194258662728,0.006,-2.555,-0.0027241378316077694
104:108,41.878938853142,12.508194258662728,0.006,-1.008,-0.0027241378316077694
104:109,41.87894874567211,12.508190635079867,0.006,3.9239998,-0.0027241378316077694
104:110,41.87894874567211,12.508190635079867,0.006,2.59,-0.0027241378316077694
104:111,41.87894514838766,12.508189427226592,0.006,4.581,-0.0027241378316077694
104:112,41.8789334572164,12.50819184294466,0.006,1.294,-0.0027241378316077694
104:113,41.8789334572164,12.50819184294466,0.006,1.843,-0.0027241378316077694
104:114,41.8789199674029,12.508193050797935,0.004,1.8199999,-0.0051034480365442815
104:115,41.87890737690945,12.508196674380796,0.003,0.962,-0.006482758543229309
104:116,41.87890018234269,12.508201505828453,0.004,0.62,-0.00586206872208879
104:117,41.87890018234269,12.508201505828453,0.004,0.754,-0.00586206872208879
104:118,41.87889568573819,12.50821600014838,0.004,-0.529,-0.00586206872208879
104:119,41.87889748438042,12.508217208001655,0.004,1.989,-0.00586206872208879
104:120,41.87889748438042,12.508217208001655,0.004,0.29,-0.00586206872208879
104:121,41.87889748438042,12.508217208001655,0.004,-1.9619999,-0.00586206872208879
104:122,41.87888489388482,12.508220831584516,0.006,-1.115,-0.004241379202696785
104:123,41.87887140407133,12.508223247302585,0.007,-0.266,-0.003620689381556265
104:124,41.87885881358218,12.508226870885448,0.011,0.006,-8.673617379884035E-19
104:125,41.87885881358218,12.508226870885448,0.011,0.122,-8.673617379884035E-19
104:126,41.87885791425784,12.508228078750241,0.011,0.267,-8.673617379884035E-19
104:127,41.87885791425784,12.508228078750241,0.011,0.39299998,-8.673617379884035E-19
104:128,41.87885791425784,12.508228078750241,0.011,0.458,-8.673617379884035E-19
104:129,41.87885791425784,12.508228078750241,0.011,0.4,-8.673617379884035E-19

End (104)

3.7. Last

Last 104

It gets the last position of the user 104

The user Id (#) must be in the range 0...255.

The Arianna Server answers with the last line of the messages of the “Get” command.

Each line contains User Id plus 5 values:

- User Id: (user Id)
- Latitude, (Latitude of the I step)
- Longitude, (Longitude of the I step)
- Zi, (Inertial Altitude)
- Zp, (Barometric Altitude)
- Zic (Inertial Altitude fused with the Barometric Altitude)

Example

Last 104

```
104:129,41.87885791425784,12.508228078750241,0.011,0.4,-8.673617379884035E-19
```

4. Other functionalities

4.1. Wayback

On a single user track can be achieved the way back path by using the command:

Wayback #, {*Len*, {*iStart*, *iEnd*}}

Parameters:

- User Id: (user Id)
- Len: it sets the length portion of the wayback track retrieved (0 for the full track).
- iStart, iEnd: index of the starting step and of the ending step of the wayback

The library calculates and returns the first "*Len*" meters of the shortest path for the # operator, between the *iStart* step and the *iEnd* step.

The Wayback command returns the sequence of points (Latitude, Longitude, Altitude) for the wayback path.

4.2. Inertial - GPS fusion

If the Arianna Data Stream contains also the running GPS fixes, the Inertial - GPS fusion can be applied.

Pay attention, the field GPS accuracy must be consistent.

TBD